

Jackson Huberty

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BioResource & Agricultural Engineering senior focused on hands-on mechanical design, prototyping, and manufacturing.

EDUCATION

California Polytechnic State University, San Luis Obispo

Expected Dec 2027

B.S. BioResource & Agricultural Engineering (BRAE) | Senior

Relevant coursework: 3D Solids Modeling, Mechanical Systems in Agriculture, Hydraulics, Engineering Surveying, Irrigation, Agricultural Structures, Fundamentals of Electricity

EXPERIENCE

Structural Lead, ASABE Quarter Scale Tractor Team | Cal Poly BRAE, San Luis Obispo, CA Aug 2026 – Present

- Lead design and fabrication across the tractor frame, steering, hydraulics, braking, fuel-system, and drivetrain-mount subsystems; coordinate interfaces with other subsystem leads.
- Modeled brake-caliper mounts, brake-steering assembly, operator platforms, fuel tank, seat mount, steering column, tow hook, and hitch mount in SOLIDWORKS for fabrication and vehicle integration.
- Resolved packaging, fit-up, and alignment issues involving frame geometry, fuel capacity, hydraulic-fitting compatibility, under-seat battery and fuel placement, engine-to-drivetrain alignment, and manufacturing variation in plasma-cut and 5-ft Hydra-bend parts.
- Fabricate and assemble structural and mechanical components using welding, CNC plasma cutting, manual milling and turning, drill presses, hydraulic assembly, and mechanical fasteners.

Senior Project Manager, Agave Harvester | Cal Poly BRAE, San Luis Obispo, CA

Sep 2026 – Present

- Lead a six-person redesign of an agave-harvester chopping head for an existing grape-harvester platform, targeting removal of approximately 80% of agave leaves for harvesting.
- Replace the prior flat-steel chopping-head structure with an square tubing design; commission the existing machine, test cutting performance, and iterate before the year-end deadline.

Shop Technician | Cal Poly BRAE Shop, San Luis Obispo, CA

Apr 2024 – Present

- Train and supervise 8-10 students per day, as needed, on safe use of fabrication equipment while enforcing PPE, machine-guarding, and shop-safety procedures.
- Maintain, clean, tram, and repair shop equipment including CNC mills, lathes, CNC plasma systems, welding equipment, and hydraulic bending equipment.
- Helped return a CNC mill to service and replace elevating-nut on Carlton drill press after failure.

Agricultural Operations Technician | Cal Poly Crops Unit, San Luis Obispo, CA

Jun 2025 – Sep 2025

- Supported crop production across approximately 8 acres of corn, sweet corn, peppers, strawberries, pumpkins, tomatoes, and broccoli using drip irrigation, traveling big gun and hand-move sprinklers.
- Operated and maintained tractors and farm implements for soil preparation, irrigation, fertilization, and harvesting.

Project Engineer Intern | Preston Companies, San Jose, CA

Jun 2024 – Sep 2024

- Coordinated 20-30 RFIs, material deliveries, and field activities across excavation, pipe-laying, concrete, and supply teams on a utility/construction work site of approximately 4 acres.
- Used 3D BIM models to identify and help resolve approximately 12 constructability conflicts involving piping, walls, foundations, and structural columns before installation.

SKILLS & CERTIFICATIONS

CAD & Design: SOLIDWORKS (CSWA, March 2024), Fusion 360, AutoCAD, mechanical product design, engineering drawings, design for manufacturing.

Software: SOLIDWORKS CAM, Fusion 360 CAM, MATLAB, Excel, Word, PowerPoint, finite element analysis coursework.

Manufacturing: Manual/CNC milling, lathe, plasma cutting, Hydra-bend, MIG/TIG/Stick welding, oxy-acetylene, G-code generation, metal fabrication and 3-D printing.

Equipment: Tractors, forklifts, excavators, reach lifts, farm machinery, hydraulics, mechanical troubleshooting.

INVOLVEMENT

- **Eagle Scout, Boy Scouts of America** - Held six leadership roles including Senior Patrol Leader; planned and coordinated a community service project.
- **Agricultural Engineering Society** - BBQ Chair (2024-2025); managed weekly event logistics and purchasing.
- **Cal Poly Tractor Pull** - Maintain competition tractors and support annual Poly Royal Tractor Pull operations.